

a-deep-dive-into-openapi-security.pdf (English translation)

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Content (translated into English)

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A DEEP DIVE INTO OPENAPI SECURITY ANDREI AGAPE – OWASP COPENHAGEN 2024 AGENDA

1. BACKGROUND
2. OPENAPI ANALYSIS
1. RESEARCH
2. TOOL
3. RESULTS
1. CONCLUSION

2

1. BACKGROUND

20XX Pitch Deck 4 5 Pitch Deck 7 8 IT'S SOMETHING .. \(\)/

9 10 CAN WE DO BETTER?

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1. OPENAPI ANALYSIS

WHAT IS OPENAPI

1. a "specification language" for REST APIs

2. describes the structure and syntax of the API

1. irrelevant of the programming language in which the

API was created

1. Previously known as SwaggerFile (v2.x) it became

OpenAPI (v3.x +)

1. has YAML or JSON format

13 WHY IS NEEDED

1. quickly discover how an API works

2. a standardized way to describe the API

1. no need to understand how the API is implemented

2. improves design/deployment/testing of API

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1. A

2. B

1. C

20XX Pitch Deck 18 19 ROLE IN DEVELOPMENT LIFECYCLE

1. Requirements

2. Design

1. Configuration

2. Development

3. Publish

1. Deployment

2. Test

20 21 ???

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1. RESEARCH

QUESTIONS

1. CAN WE USE THE OPENAPI DOCS TO FIND DESIGN FAULTS?

2. WHAT TYPE OF VULN CLASSES ARE THESE?

1. IDEAS AND EXAMPLES?

24

1. INDIRECT BROKEN

ACCESS CONTROL

Looks fine $\neg()/\neg$

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1. INDIRECT BROKEN

ACCESS CONTROL

1. FIND OBJECT WITH SENSITIVE INFO
2. CHECK ALL PATHS TO READ/MODIFY IT
1. CHECK FOR DIRECT AND INDIRECT ACCESS

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1. INCONSISTENT DESIGN

VULNERABLE?

27

1. INCONSISTENT DESIGN

VULNERABLE?

28

1. INCONSISTENT DESIGN
1. CHECK ENDPOINT'S INPUT/OUTPUT
2. ANALYZE ENDPOINT'S DESCRIPTION
1. DETERMINE IF THEY MATCH

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1. IDOR ID FINDER
1. UUID might be used for access control
2. But how to check?

30

1. IDOR ID FINDER
1. FIND ENDPOINTS THAT REQUIRE ID
2. DETERMINE WHERE THESE IDS ARE LEAKED
1. OBTAIN THEM AND TEST ACCESS CONTROL

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1. THE TOOL

CHARACTERISTICS

1. WORK ON ANY OPENAPI REGARDLESS OF THE API STRUCTURE
2. VISUAL + SCRIPTING
1. EASY CONFIGURABLE

33 OPTIONS

1. JSON SCHEMA-BASED OBJECT NAVIGATION
2. GRAPH TRAVERSAL ALGORITHMS

1. GRAPH DATABASE LIKE NEO4J

34 35 CONCEPT

1. SIMILAR TO BLOODHOUND, BUT FOR REST API
2. ENDPOINTS, PARAMETERS, OBJECTS, ETC. -> NODES

1. FIND VULNS USING NEO4J QUERIES AND PATHS

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1. RESULTS

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1. INDIRECT BROKEN

ACCESS CONTROL

1. FIND OBJECT WITH SENSITIVE INFO
2. CHECK ALL PATHS TO READ/MODIFY IT
1. CHECK FOR DIRECT AND INDIRECT ACCESS

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1. PATHS TO SENSITIVE DATA

1. LOOK FOR FIELDS/OBJECT CONTAINING SENSITIVE DATA
2. CHECK ALL PUBLIC PATHS TO REACH IT
1. EXFIL THE DATA THROUGH UNPROTECTED ENDPOINTS

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1. BYPASS RATE LIMIT

1. SET THE TARGET FIELD/OBJECT
2. LIST ALL PATHS TO REACH IT
1. BYPASS RATE LIMIT BY PARALELIZING THE REQUESTS

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1. IDOR ID FINDER

1. FIND ENDPOINTS THAT REQUIRE ID
2. DETERMINE WHERE THESE IDS ARE LEAKED
1. OBTAIN THEM AND TEST ACCESS CONTROL

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1. IDOR ID

FINDER

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1. CONCLUSION

CONCLUSION

1. PARSING OPENAPI IS A COMPLEX TASK
2. POC TOOL IS BUGGY, BUT HAS POTENTIAL
1. NEO4J QUERIES CAN DETECT POSSIBLE INSECURE DESIGN
2. MANUAL VERIFICATION IS (STILL) NEEDED
3. PROMISING TECHNIQUE TO IMPROVE DEFENSE/ATTACK OF API

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